

DUKE ODOO to FieldKonnnect

INBOUND API REQUIREMENTS AND DATA CONTRACT

Prepared for

Duke Odoo Team and FieldKonnnect Team

Integration direction

Duke Odoo to FieldKonnnect

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CONFIDENTIAL

For review and approval by authorized Duke and FieldKonnnect stakeholders.

Inbound API Requirements and Data Contract

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1. Purpose

This document defines the APIs and data fields required from Duke Odoo to synchronize the following data into FieldKonnnect:

1. Distributor/Customer Master (create and update)
2. Party-wise Product Pricing
3. Product Category Master
4. Product Sub-category Master
5. Product Master
6. Employee Leave Balance
7. Order Status and Dispatch Updates
8. Invoice Data
9. Customer Outstanding Data

It is intended to be the working contract between the Duke Odoo API team and the FieldKonnnect integration team. Endpoint paths in this document are recommended paths; the final URLs may be changed by mutual agreement, but the request/response semantics and stable business keys must remain consistent.

2. Integration Principles

2.1 Source of truth

- Duke Odoo will be the source of truth for all modules covered by this document.
- FieldKonnnect will consume the Odoo APIs and upsert records using Odoo-provided stable external identifiers.
- FieldKonnnect-generated numeric database IDs must never be used by Odoo as integration keys.
- A record omitted from a response must not be treated as deleted. Odoo must explicitly send active: false, is_deleted: true, or a separately agreed deletion event.

2.2 Recommended synchronization model

- Initial sync: Full paginated pull of all active and inactive records.
- Incremental sync: Pull records changed after updated_since, ordered by updated_at and external_id.
- Frequency: Master data every 15-30 minutes; order/dispatch, invoice and outstanding data every 5-15 minutes, subject to business approval.
- Timezone: ISO 8601 timestamps with offset. UTC is preferred, for example 2026-08-27T10:30:00Z.
- Date format: YYYY-MM-DD.
- Decimal format: JSON number using a dot as decimal separator; no commas or currency symbols.
- Null handling: Optional unavailable values must be null, not an empty string.

2.3 Dependency order

S.No.	Data	Must be synchronized after
1	Product Category	None

S.No.	Data	Must be synchronized after
2	Product Sub-category	Product Category
3	Product Master	Category and Sub-category
4	Distributor/Customer Master	Supporting geography masters, if codes are used
5	Party-wise Product Pricing	Party and Product Master
6	Leave Balance	Employee/User mapping
7	Order/Dispatch Updates	FieldKonnnect order and Product Master
8	Invoice Data	Party, Product and preferably Order mapping
9	Customer Outstanding	Party Master

3. Common API Standards

3.1 Authentication and transport

Odoo must provide:

- If Odoo provides a publicly accessible API URL, separate API authentication is not required.
- If the API is not publicly accessible, Odoo must provide the authentication method and credentials (OAuth 2.0, API key/HMAC, or another mutually agreed method).
- All endpoints must use HTTPS (TLS 1.2 or later), whether public or authenticated.
- Separate UAT and Production URLs; separate credentials where authentication applies.
- Credentials must never be placed in query parameters.

Recommended headers:

```
Authorization: Bearer <token> (only when authentication is required)
Accept: application/json
Content-Type: application/json
X-Correlation-ID: <uuid>
X-API-Version: 1
```

3.2 Common query parameters for list APIs

S.No.	Parameter	Type	Requirement	Description
1	updated_since	ISO 8601 datetime	Optional for full sync; required for delta sync	Return records updated strictly after this cursor/time
2	updated_until	ISO 8601 datetime	Optional	Creates a stable synchronization window
3	page	Integer	Required	Page number, starting from 1
4	page_size	Integer	Required	Recommended 100; maximum at least 500
5	include_inactive	Boolean	Optional	Include inactive/deactivated records
6	external_id	String	Optional	Fetch or reconcile one Odoo record

3.3 Standard success response

```
{
  "success": true,
  "message": "Records fetched successfully",
  "data": [],
  "pagination": {
    "page": 1,
    "page_size": 100,
    "total_records": 245,
    "total_pages": 3,
    "has_next": true
  },
  "sync": {
    "generated_at": "2026-08-27T10:30:00Z",
    "next_cursor": "2026-08-27T10:29:59.456Z"
  },
  "correlation_id": "89af1ec8-c068-42fc-99ca-03a69a988265"
}
```

3.4 Standard error response

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "One or more fields are invalid",
    "details": [
      {
        "field": "party_code",
        "message": "party_code is required"
      }
    ]
  },
  "correlation_id": "89af1ec8-c068-42fc-99ca-03a69a988265"
}
```

Required status codes: 200, 400, 401, 403, 404, 409, 422, 429, and 500/503. Odoo must document rate limits and return Retry-After with 429 or temporary 503 responses.

3.5 Common fields on every synchronized record

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo technical ID; primary integration key
2	created_at	ISO 8601 datetime	Required	Record creation time in Odoo
3	updated_at	ISO 8601 datetime	Required	Last functional modification time
4	active	Boolean	Required	Whether the record is active
5	is_deleted	Boolean	Optional, default false	Explicit logical deletion indicator
6	company_code	String(50)	Required if Odoo is multi-company	Duke legal company/business unit code

external_id must never change after creation. Business codes may be editable only if Odoo also continues to send the immutable external_id.

4. API Summary

S.No.	Recommended endpoint	Method	Purpose	Expected mode
1	/api/v1/fieldconnect/parties	GET	Distributor/customer create and update feed	Full + delta
2	/api/v1/fieldconnect/party-prices	GET	Party-specific product prices	Full + delta

S.No.	Recommended endpoint	Method	Purpose	Expected mode
3	/api/v1/fieldconnect/product-categories	GET	Product category create/update feed	Full + delta
4	/api/v1/fieldconnect/product-subcategories	GET	Product sub-category create/update feed	Full + delta
5	/api/v1/fieldconnect/products	GET	Product and product-variant master	Full + delta
6	/api/v1/fieldconnect/leave-balances	GET	Employee leave balances	Full + delta
7	/api/v1/fieldconnect/order-dispatches	GET	Order status and dispatch updates	Delta
8	/api/v1/fieldconnect/invoices	GET	Invoice header and line data	Full + delta
9	/api/v1/fieldconnect/customer-outstandings	GET	Party receivable/outstanding data	Snapshot + delta

5. API 1 - Distributor/Customer Master

Endpoint: GET /api/v1/fieldconnect/parties Purpose: Send new and updated distributors, dealers and other customer/party records from Odoo.

5.1 Party fields

S.No.	Field	Type	Requirement	FieldConnect use / validation
1	external_id	String(100)	Required	Immutable Odoo key
2	party_code	String(100)	Required	Odoo/SAP customer or distributor code; unique per company
3	party_type	Enum	Required	distributor, dealer, retailer, customer, service_center, or agreed master value
4	legal_name	String(255)	Required	Legal/registered name
5	trade_name	String(255)	Optional	Display/shop name; falls back to legal name
6	business_status	Enum	Required	Recommended: active, blocked, on_hold, closed
7	business_start_date	Date	Optional	Date business relationship started
8	contact_person	String(150)	Optional	Primary contact
9	contact_designation	String(100)	Optional	Contact's designation
10	mobile	String(20)	Required	Include country code; digits/leading + only
11	alternate_mobile	String(20)	Optional	Alternate contact
12	email	String(255)	Optional	Valid email
13	secondary_email	String(255)	Optional	Secondary email
14	gst_number	String(20)	Conditional	Required for GST-registered parties
15	pan_number	String(20)	Optional	PAN number
16	registration_type	String(50)	Optional	GST/registration classification
17	customer_segment	String(100)	Optional	Odoo customer segmentation
18	credit_limit	Decimal(15,2)	Optional	Non-negative
19	credit_days	Integer	Optional	Non-negative
20	payment_term_code	String(50)	Optional	Stable Odoo payment-term code

S.No.	Field	Type	Requirement	FieldKonnnect use / validation
21	sales_zone_code	String(50)	Optional	Prefer code over display name
22	sales_zone_name	String(150)	Optional	Display value
23	area_territory_code	String(50)	Optional	Stable territory key
24	area_territory_name	String(150)	Optional	Display value
25	beat_route_code	String(50)	Optional	FieldKonnnect beat/route mapping key
26	sales_employee_code	String(50)	Optional	Must match FieldKonnnect users.employee_codes
27	supervisor_employee_code	String(50)	Optional	Reporting user mapping
28	latitude	Decimal(10,7)	Optional	Range -90 to 90
29	longitude	Decimal(10,7)	Optional	Range -180 to 180
30	billing_address	Object	Required	See address fields below
31	shipping_address	Object	Optional	May be omitted when same_as_billing=true
32	same_as_billing	Boolean	Required	Determines shipping fallback
33	bank_details	Object	Optional	Bank name, account holder/no., IFSC, branch
34	active	Boolean	Required	Maps to active/inactive state
35	is_deleted	Boolean	Optional	Logical removal only
36	created_at	Datetime	Required	Odoo creation timestamp
37	updated_at	Datetime	Required	Delta cursor field

5.2 Address object

S.No.	Field	Type	Requirement
1	address_line_1	String(255)	Required
2	address_line_2	String(255)	Optional
3	landmark	String(150)	Optional
4	city_code	String(50)	Preferred
5	city_name	String(100)	Required
6	district_code	String(50)	Preferred
7	district_name	String(100)	Required
8	state_code	String(50)	Required
9	state_name	String(100)	Required
10	country_code	ISO 3166-1 alpha-2	Required
11	country_name	String(100)	Optional
12	pincode	String(10)	Required for Indian addresses

5.3 Example party record

```
{
  "external_id": "res.partner:8421",
  "party_code": "D100245",
  "party_type": "distributor",
  "legal_name": "Example Distributors Private Limited",
  "trade_name": "Example Distributors",
  "business_status": "active",
  "contact_person": "Amit Kumar",
  "mobile": "+919876543210",
  "email": "accounts@example.com",
  "gst_number": "24ABCDE1234F1Z5",
  "credit_limit": 500000.00,
  "credit_days": 30,
  "sales_zone_code": "WEST",
  "sales_employee_code": "42610",
  "billing_address": {
    "address_line_1": "Plot 10, Industrial Area",
    "city_name": "Ahmedabad",
    "district_name": "Ahmedabad",
    "state_code": "GJ",
    "state_name": "Gujarat",
    "country_code": "IN",
    "pincode": "380001"
  },
  "same_as_billing": true,
  "active": true,
  "is_deleted": false,
  "created_at": "2025-04-01T09:00:00Z",
  "updated_at": "2026-08-27T10:20:00Z"
}
```

5.4 Current FieldKonnnect mapping notes

- FieldKonnnect currently has both customers and master_distributors; party_type must determine the target entity.
- party_code maps to customers.sap_code/customer_code or master_distributors.distributor_code as agreed.
- Current FieldKonnnect customer mobile and email fields have uniqueness constraints. Duplicate handling must be finalized before production sync.
- Odoo location codes are preferred. Name-only matching can create duplicate cities/states due to spelling differences.

6. API 2 - Party-wise Product Pricing

Endpoint: GET /api/v1/fieldkonnnect/party-prices Purpose: Supply the effective product price for a specific customer/distributor and validity period.

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo price-rule ID
2	price_list_code	String(50)	Required	Odoo pricelist code
3	price_list_name	String(150)	Optional	Display name
4	party_external_id	String(100)	Required	Party immutable Odoo ID
5	party_code	String(100)	Required	Reconciliation key
6	product_external_id	String(100)	Required	Product/variant Odoo ID
7	product_code	String(100)	Required	Must match Product Master
8	currency_code	ISO 4217	Required	Example: INR
9	base_price	Decimal(19,4)	Required	Price before party rule
10	party_price	Decimal(19,4)	Required	Final price before tax unless tax_inclusive=true
11	discount_percent	Decimal(8,4)	Optional	Non-negative

S.No.	Field	Type	Requirement	Description
12	tax_inclusive	Boolean	Required	Price tax treatment
13	minimum_quantity	Decimal(19,4)	Optional, default 1	Slab minimum
14	maximum_quantity	Decimal(19,4)	Optional	null means no maximum
15	uom_code	String(50)	Required	Unit of measure for this price
16	valid_from	Datetime	Required	Effective start
17	valid_to	Datetime	Optional	null means open-ended
18	priority	Integer	Optional, default 0	Resolves overlapping rules; higher wins
19	active	Boolean	Required	Rule state
20	updated_at	Datetime	Required	Delta sync

Pricing precedence must be confirmed: exact party rule to party group/segment rule to standard product price. Odoo must not return multiple equally prioritized active rules for the same party, product, quantity and timestamp.

FieldKonnct gap: the audited application has general product pricing in product_details, but no clear dedicated party-wise price entity. A new integration table with unique Odoo key, party, product, validity, quantity slab and priority will be required before this API is implemented.

7. API 3 - Product Category Master

Endpoint: GET /api/v1/fieldkonnct/product-categories

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo category ID
2	category_code	String(100)	Required	Unique business/SAP code
3	category_name	String(250)	Required	Display name
4	description	String/Text	Optional	Category description
5	ranking	Integer	Optional, default 1	Display order
6	image_url	URL	Optional	HTTPS URL accessible to FieldKonnct
7	active	Boolean	Required	Active/inactive state
8	is_deleted	Boolean	Optional	Logical deletion
9	created_at	Datetime	Required	Creation time
10	updated_at	Datetime	Required	Delta sync

FieldKonnct mapping: category_name, ranking, category_image, sap_code, and active. Category code must be unique and is the preferred relationship key for sub-categories.

8. API 4 - Product Sub-category Master

Endpoint: GET /api/v1/fieldkonnct/product-subcategories

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo sub-category ID
2	subcategory_code	String(100)	Required	Unique business/SAP code
3	subcategory_name	String(250)	Required	Display name

S.No.	Field	Type	Requirement	Description
4	category_external_id	String(100)	Required	Parent category Odoo ID
5	category_code	String(100)	Required	Parent reconciliation key
6	description	String/Text	Optional	Description
7	ranking	Integer	Optional, default 1	Display order
8	image_url	URL	Optional	HTTPS image URL
9	active	Boolean	Required	Active/inactive state
10	is_deleted	Boolean	Optional	Logical deletion
11	created_at	Datetime	Required	Creation time
12	updated_at	Datetime	Required	Delta sync

Odoo must send the parent category on every sub-category record. A sub-category whose parent category is unknown in FieldKonnect must be rejected into the integration error log and retried after its parent is synchronized.

9. API 5 - Product Master

Endpoint: GET /api/v1/fieldkonnect/products Purpose: Synchronize products and orderable variants.

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo product/variant ID
2	product_code	String(100)	Required	Unique order/invoice line code
3	product_name	String(250)	Required	Product name
4	display_name	String(250)	Optional	Mobile/web display name
5	description	Text	Optional	Product description
6	category_external_id	String(100)	Required	Category Odoo ID
7	category_code	String(100)	Required	Category reconciliation key
8	subcategory_external_id	String(100)	Conditional	Required if Duke uses sub-categories
9	subcategory_code	String(100)	Conditional	Required if Duke uses sub-categories
10	brand_code	String(100)	Optional	Stable brand code
11	brand_name	String(150)	Optional	Brand display name
12	uom_code	String(50)	Required	Base unit of measure
13	uom_name	String(100)	Optional	Unit display name
14	hsn_code	String(50)	Conditional	Required for taxable goods where applicable
15	ean_barcode	String(100)	Optional	EAN/barcode; unique when supplied
16	mrp	Decimal(19,4)	Required	Non-negative
17	standard_price	Decimal(19,4)	Required	Default/base selling price
18	gst_percent	Decimal(8,4)	Required	GST rate
19	currency_code	ISO 4217	Required	Example INR
20	model_no	String(250)	Optional	Model identifier
21	part_no	String(250)	Optional	Part number
22	product_group	String(150)	Optional	Maps to group/new group where agreed
23	product_subgroup	String(150)	Optional	Maps to subgroup

S.No.	Field	Type	Requirement	Description
24	specification	Text	Optional	Technical specification
25	image_url	URL	Optional	Product image URL
26	orderable	Boolean	Required	Whether FieldKonnnect may place orders
27	active	Boolean	Required	Master state
28	is_deleted	Boolean	Optional	Logical deletion
29	created_at	Datetime	Required	Creation time
30	updated_at	Datetime	Required	Delta sync

9.1 Variant rule

If Odoo maintains product templates and variants, Odoo must return one record per orderable variant. external_id, product_code, barcode and pricing must identify the variant, not only the template.

9.2 Example product record

```
{
  "external_id": "product.product:15580",
  "product_code": "PUMP-5HP-001",
  "product_name": "Duke 5 HP Pump",
  "category_external_id": "product.category:10",
  "category_code": "PUMP",
  "subcategory_external_id": "product.category:14",
  "subcategory_code": "CENTRIFUGAL",
  "uom_code": "NOS",
  "hsn_code": "84137010",
  "mrp": 25000.00,
  "standard_price": 22000.00,
  "gst_percent": 18.00,
  "currency_code": "INR",
  "orderable": true,
  "active": true,
  "created_at": "2025-01-10T08:00:00Z",
  "updated_at": "2026-08-27T10:00:00Z"
}
```

10. API 6 - Employee Leave Balance

Endpoint: GET /api/v1/fieldkonnnect/leave-balances Purpose: Synchronize current leave entitlement/balance by employee and leave type.

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Unique Odoo employee + leave type + period record ID
2	employee_external_id	String(100)	Required	Immutable Odoo employee ID
3	employee_code	String(50)	Required	Must match FieldKonnnect users.employee_codes
4	leave_type_code	String(50)	Required	Stable leave type code, e.g. CL, SL, EL, COMP_OFF
5	leave_type_name	String(100)	Required	Display name
6	period_start	Date	Required	Balance period start
7	period_end	Date	Required	Balance period end
8	opening_balance	Decimal(8,2)	Optional	Opening entitlement
9	accrued	Decimal(8,2)	Optional	Accrued in period
10	availed	Decimal(8,2)	Optional	Approved/consumed leave
11	pending	Decimal(8,2)	Optional	Pending approval amount

S.No.	Field	Type	Requirement	Description
12	adjusted	Decimal(8,2)	Optional	Manual adjustment
13	available_balance	Decimal(8,2)	Required	Authoritative balance available to apply
14	as_of_date	Date	Required	Business date of balance
15	expiry_date	Date	Optional	Particularly for comp-off
16	active	Boolean	Required	Leave type/balance active state
17	updated_at	Datetime	Required	Delta sync

10.1 Leave mapping confirmation required

FieldKconnect currently contains separate balance concepts such as casual, sick, earned leave and comp-off. Duke HR/Odoo must provide the authoritative list of leave type codes and confirm:

- Whether pending applications reduce available_balance.
- Whether Odoo or FieldKconnect is the final approval/source-of-truth system.
- Whether balances may be fractional (0.5 day).
- Carry-forward and expiry rules.
- Whether FieldKconnect should overwrite balances or post adjustments.

Recommended behavior: Odoo sends an authoritative snapshot and FieldKconnect overwrites the mapped balance for the employee and leave type after validating as_of_date is not older than the last accepted snapshot.

11. API 7 - Order Status and Dispatch Updates

Endpoint: GET /api/v1/fieldkconnect/order-dispatches Purpose: Return Odoo processing, dispatch and invoice references against orders originating in FieldKconnect.

11.1 Order header fields

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo sales order ID
2	fieldkconnect_order_id	Integer/String	Conditional	Required if sent to Odoo during order creation
3	fieldkconnect_order_no	String(250)	Required	FieldKconnect orderno; primary reconciliation reference
4	odoo_order_no	String(100)	Required	Odoo sales order number
5	order_status_code	String(50)	Required	Stable status code
6	order_status_name	String(100)	Required	Display status
7	status_reason	String(500)	Conditional	Required for rejected/cancelled/on-hold orders
8	total_order_qty	Decimal(19,4)	Required	Odoo-confirmed ordered quantity
9	total_dispatched_qty	Decimal(19,4)	Required	Cumulative dispatched quantity
10	pending_qty	Decimal(19,4)	Required	Remaining quantity
11	fully_dispatched	Boolean	Required	Header completion flag
12	cancelled_at	Datetime	Conditional	For cancellation
13	updated_at	Datetime	Required	Delta sync

11.2 Dispatch object

An order may have multiple dispatches. Odoo must send a dispatches array.

S.No.	Field	Type	Requirement	Description
1	dispatch_external_id	String(100)	Required	Immutable stock-picking/delivery ID
2	dispatch_no	String(100)	Required	Delivery/dispatch number
3	dispatch_status_code	String(50)	Required	Stable status code
4	dispatch_status_name	String(100)	Required	Display status
5	dispatch_date	Datetime	Required when dispatched	Actual dispatch time
6	transporter_code	String(50)	Optional	Stable transporter ID
7	transporter_name	String(150)	Optional	Transporter display name
8	vehicle_no	String(50)	Optional	Vehicle number
9	lr_no	String(100)	Conditional	Required when LR is generated
10	lr_date	Date	Optional	LR date
11	tracking_url	URL	Optional	Shipment tracking URL
12	invoice_no	String(100)	Optional	Linked invoice number
13	invoice_date	Date	Conditional	Required when invoice number is supplied
14	lines	Array	Required	Product-level dispatch quantities

11.3 Dispatch line fields

S.No.	Field	Type	Requirement
1	line_external_id	String(100)	Required
2	fieldkonnect_order_line_id	Integer/String	Optional but strongly recommended
3	product_external_id	String(100)	Required
4	product_code	String(100)	Required
5	ordered_qty	Decimal(19,4)	Required
6	dispatch_qty	Decimal(19,4)	Required
7	cumulative_dispatched_qty	Decimal(19,4)	Required
8	pending_qty	Decimal(19,4)	Required
9	uom_code	String(50)	Required
10	line_status_code	String(50)	Required

11.4 Mandatory status mapping

Before UAT, Odoo must provide the full status-code catalogue and mapping to at least:

received, confirmed, processing, on_hold, partially_dispatched, fully_dispatched, invoiced, rejected, and cancelled.

FieldKonnnect already stores order and order-line shipped quantities and status references. Updates must be idempotent and cumulative values must never decrease unless Odoo sends an explicit correction event with a reason.

12. API 8 - Invoice Data

Endpoint: GET /api/v1/fieldkonnect/invoices Purpose: Synchronize finalized invoices and credit/cancellation state.

12.1 Invoice header fields

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo invoice ID
2	invoice_no	String(100)	Required	Unique within company/fiscal year
3	invoice_type	Enum	Required	invoice, credit_note, debit_note
4	invoice_date	Date	Required	Posting/invoice date
5	posting_date	Date	Optional	Accounting posting date
6	due_date	Date	Optional	Payment due date
7	fiscal_year	String(20)	Required	Example 2026-27
8	company_code	String(50)	Required	Legal entity
9	party_external_id	String(100)	Required	Customer/distributor Odoo ID
10	party_code	String(100)	Required	Party reconciliation key
11	fieldkconnect_order_no	String(250)	Optional but strongly recommended	Source order reference
12	odoo_order_no	String(100)	Optional	Odoo sales order reference
13	dispatch_external_id	String(100)	Optional	Delivery reference
14	currency_code	ISO 4217	Required	Invoice currency
15	subtotal_amount	Decimal(19,2)	Required	Before tax
16	discount_amount	Decimal(19,2)	Optional, default 0	Total discount
17	tax_amount	Decimal(19,2)	Required	Total tax
18	round_off_amount	Decimal(19,2)	Optional, default 0	Round-off
19	total_amount	Decimal(19,2)	Required	Invoice total
20	paid_amount	Decimal(19,2)	Optional	Amount received
21	balance_amount	Decimal(19,2)	Required	Outstanding on this invoice
22	payment_status	Enum	Required	unpaid, partially_paid, paid, overdue
23	invoice_status	Enum	Required	draft, posted, cancelled, reversed
24	pdf_url	URL	Optional	Authenticated or time-limited invoice PDF URL
25	irn	String(100)	Optional	E-invoice IRN
26	ack_no	String(100)	Optional	E-invoice acknowledgement
27	eway_bill_no	String(100)	Optional	E-way bill
28	lines	Array	Required	Invoice line items
29	created_at	Datetime	Required	Creation time
30	updated_at	Datetime	Required	Delta sync

12.2 Invoice line fields

S.No.	Field	Type	Requirement
1	line_external_id	String(100)	Required
2	product_external_id	String(100)	Required for product lines
3	product_code	String(100)	Required for product lines
4	description	String(500)	Required

S.No.	Field	Type	Requirement
5	quantity	Decimal(19,4)	Required
6	uom_code	String(50)	Required
7	unit_price	Decimal(19,4)	Required
8	discount_percent	Decimal(8,4)	Optional
9	discount_amount	Decimal(19,2)	Optional
10	hsn_code	String(50)	Conditional
11	gst_percent	Decimal(8,4)	Required
12	cgst_amount	Decimal(19,2)	Conditional
13	sgst_amount	Decimal(19,2)	Conditional
14	igst_amount	Decimal(19,2)	Conditional
15	tax_amount	Decimal(19,2)	Required
16	line_total	Decimal(19,2)	Required

FieldKconnect currently contains sales/sales_details, primary_sales, and separate invoice-related entities. The business team must confirm which FieldKconnect invoice screen/report is authoritative before implementation. The Odoo invoice API must retain order, party and product codes to support reconciliation regardless of the final storage entity.

13. API 9 - Customer Outstanding Data

Endpoint: GET /api/v1/fieldkconnect/customer-outstandings Purpose: Provide current receivables/outstanding balances and ageing details.

13.1 Recommended response granularity

Odoo should return invoice-level open items plus a party-level total. Invoice-level data enables ageing, reconciliation and drill-down; a single party balance alone is insufficient for reliable reporting.

S.No.	Field	Type	Requirement	Description
1	external_id	String(100)	Required	Immutable Odoo open-item ID
2	party_external_id	String(100)	Required	Odoo party ID
3	party_code	String(100)	Required	Party reconciliation key
4	party_name	String(255)	Required	Display/reconciliation value
5	company_code	String(50)	Required	Legal entity
6	invoice_external_id	String(100)	Conditional	Required for invoice-level item
7	invoice_no	String(100)	Conditional	Required for invoice-level item
8	invoice_date	Date	Conditional	Required for invoice-level item
9	due_date	Date	Conditional	Required for ageing
10	currency_code	ISO 4217	Required	Currency
11	original_amount	Decimal(19,2)	Conditional	Invoice original amount
12	paid_amount	Decimal(19,2)	Optional, default 0	Amount settled
13	credit_note_amount	Decimal(19,2)	Optional, default 0	Credit adjustment
14	outstanding_amount	Decimal(19,2)	Required	Authoritative open amount
15	overdue_amount	Decimal(19,2)	Required	Amount past due

S.No.	Field	Type	Requirement	Description
16	days_overdue	Integer	Required	0 when not overdue
17	ageing_bucket	Enum	Required	Agreed values, e.g. CURRENT, 1_30, 31_60, 61_90, 91_180, 180_PLUS
18	credit_limit	Decimal(19,2)	Optional	Current party limit
19	available_credit	Decimal(19,2)	Optional	Credit limit minus utilized credit
20	as_of_date	Date	Required	Snapshot business date
21	updated_at	Datetime	Required	Delta sync

13.2 Snapshot rules

- Odoo must state whether the endpoint is a complete snapshot or changed open items only.
- For a complete snapshot, the response must include snapshot_id, as_of_date, and is_complete: true so FieldKconnect can safely close items absent from the new snapshot.
- For delta mode, settlement/closure must be explicitly sent with outstanding_amount: 0 and a closed status.
- Negative balances/advances must be supported and documented rather than converted to zero.

FieldKconnect note: an existing customer_outstandings entity stores party, amount and days but is not sufficient for robust invoice-level ageing. Additional external ID, invoice reference, currency, due date, ageing, snapshot and status fields are recommended.

14. Idempotency, Upsert and Conflict Rules

1. FieldKconnect will upsert using company_code + external_id.
2. Re-sending the same record must not create duplicates.
3. An update with an older updated_at than the last accepted version must be ignored and logged.
4. Odoo must return all currently valid child lines for invoice/dispatch replacement semantics, or provide immutable line events. The selected behavior must be documented.
5. Business code changes must be sent as updates against the same immutable external_id.
6. Deletes will be logical only. Physical deletion is outside this integration.
7. Failed records must not block valid records in the same page; FieldKconnect will record per-item failures and retry them.

15. Reconciliation Requirements

Odoo must provide either dedicated count/checksum endpoints or the following metadata for every list API:

- Record count for requested filters/window.
- Minimum and maximum updated_at in the response.
- Stable next_cursor.
- Unique response/correlation ID.
- Full-sync completion indicator.

Recommended reconciliation endpoint:

```
GET /api/v1/fieldkconnect/reconciliation?entity=products&as_of=2026-08-27
```

Recommended response fields: entity, company_code, active_count, inactive_count, latest_updated_at, and optional checksum.

16. Security, Audit and Data Retention

- Log authentication client, request time, endpoint, filters, response status and correlation ID.
- Do not log access tokens, bank account numbers, PAN, or full personal contact data in plain text.
- Encrypt credentials and sensitive payload archives at rest.
- Maintain an integration audit log in FieldKonnnect containing entity, external ID, received version, result and error message.
- Agree retention periods for raw payloads and retry/dead-letter records.
- Production data must not be used in UAT unless masked and approved.

17. Non-functional Requirements

S.No.	Requirement	Expected standard
1	Availability	At least 99.5% during agreed business/integration window
2	Response time	95th percentile under 5 seconds for a page of 100 records
3	Pagination	Deterministic; no duplicates/missing records while traversing a fixed window
4	Rate limit	Must be documented; minimum recommended 60 requests/minute
5	Payload size	Maximum response size must be documented
6	Retry safety	All GETs and repeated pages must be idempotent
7	Versioning	Breaking changes only under a new version such as /v2/
8	Monitoring	Odoo and FieldKonnnect teams must monitor failures and stale syncs

18. UAT Acceptance Criteria

The integration will be accepted only after the following scenarios pass:

1. Initial full sync and subsequent delta sync for each master.
2. Create, update, deactivate and reactivate scenarios.
3. Duplicate delivery of the same record without duplicate insertion.
4. Category to sub-category to product dependency handling.
5. Party and product code change using the same external ID.
6. Party pricing with validity periods and quantity slabs.
7. Employee not found and leave type not mapped scenarios.
8. Partial dispatch across multiple dispatches and products.
9. Fully dispatched, rejected and cancelled order scenarios.
10. Multiple invoices against one order and one invoice covering multiple dispatches, if supported.
11. Invoice correction/credit note scenario.
12. Outstanding invoice becoming partially paid and then fully settled.
13. Pagination over at least 1,000 records with no duplicates or omissions.
14. API timeout, 429, 500 and retry behavior.

15.Reconciliation counts matching between Odoo and FieldKonnnect.

19. Decisions Required Before Development

S.No.	Open decision	Owner
1	Final Odoo base URLs for UAT and Production	Duke Odoo Team
2	Authentication method if API URL is private	Duke IT/Odoo Team
3	Pull schedule vs Odoo webhook/push model	Both teams
4	Final party types and mapping to customers vs master_distributors	Duke Business + FieldKonnnect
5	Duplicate mobile/email policy for party records	Duke Business
6	Geography mapping by codes or names	Both teams
7	Party pricing precedence and tax inclusion	Duke Odoo/Business Team
8	Product template vs variant granularity	Duke Odoo Team
9	Complete leave type mapping and authority for approvals	Duke HR
10	Order status catalogue and mapping	Duke Odoo/Business Team
11	FieldKonnnect order handoff key available in Odoo	Both teams
12	Authoritative FieldKonnnect storage/screen for Odoo invoices	FieldKonnnect + Duke Business
13	Outstanding feed: invoice-level snapshot or delta events	Duke Finance/Odoo Team
14	Data retention, retry and reconciliation SLA	Duke IT + FieldKonnnect

20. FieldKonnnect Implementation Impact Summary

Based on the current project audit:

- Existing entities can support core category, sub-category, product, order, order-line, sales/invoice, customer, distributor, user leave balance and basic outstanding data.
- Stable Odoo external_id and synchronization metadata are not consistently available across these entities; integration mapping fields/tables will be required.
- A dedicated party-wise pricing structure is required.
- Customer outstanding storage should be extended for invoice-level ageing and snapshot reconciliation.
- Distributor/customer unification and type mapping must be decided before implementation.
- Order status IDs must not be accepted directly from Odoo; Odoo status codes must be mapped to FieldKonnnect status IDs.
- Product, party, employee and geography relationships must use stable codes/external IDs, not environment-specific numeric IDs.
- Every importer should use database transactions, validation, idempotent upserts, audit logging and retry/dead-letter handling.

21. Sign-off

Role	Name	Signature/Approval	Date
Duke Business Owner			

Role	Name	Signature/Approval	Date
Duke Odoo Technical Owner			
Duke Finance R epresenta tive			
Duke HR R epresenta tive			
FieldKonn ect Technical Owner			

End of document